

Kabadiwala Connect - PRD

Product Requirements Document

Kabadiwala Connect — Formal Recycling Chain Integration Platform

SIH26229 | Ministry of Mines (MoM)

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1. Problem Statement

India's end-of-life electronics (e-waste) are predominantly collected through informal scrap dealers, waste-pickers, and local aggregators due to their extensive last-mile reach. However, these informal collectors remain largely outside the formal recycling ecosystem established under the **E-Waste (Management) Rules, 2022**, which mandates an Extended Producer Responsibility (EPR) framework involving CPCB-registered recyclers.

As a result, materials collected informally often undergo unsafe backyard processing — open-air cable burning, acid leaching of PCBs, manual desoldering without extraction facilities — causing loss of recoverable high-value materials (lithium, cobalt, neodymium, tantalum, gallium, indium) and significant occupational health risk. The core gap is informational and institutional, not purely technological: collectors lack visibility into fair market pricing, authorized recycler networks, and compliant handover documentation.

2. Objective

Design and deploy a role-based digital platform that makes participation in the formal recycling chain the economically rational choice for informal collectors — reducing friction, increasing price transparency, and generating a traceable chain-of-custody from source to authorized processor.

3. Research & Literature Review

This section documents the research base informing the platform design — the studies reviewed, the data points drawn from them, and how each finding maps to a specific design decision in this platform.

3.1 Papers & Sources Reviewed

1. **Sengupta et al., “Circular economy and household e-waste management in India, Part II: A case study on informal e-waste collectors (Kabadiwalas) in India”** (*Resources, Conservation and Recycling*, ScienceDirect, 2023)
 - GPS-tracked kabadiwala collection routes across Indian cities
 - Finding: kabadiwalas travel 20–30 km/day using manual/three-wheeler transport; e-waste constitutes ~10% of total waste collected per collector
 - Finding: ~45% of surveyed households prefer disposing e-waste to kabadiwalas over any formal channel
 - Conclusion drawn by authors: integrating kabadiwalas into the formal system (rather than bypassing them) is necessary for effective e-waste recovery in India
2. **Study on informal e-waste ecosystem, Chandigarh** (semi-structured stakeholder interviews, published via ResearchGate)
 - Identifies deficiencies in EPR policy enforcement at the ground level
 - Documents **widespread circulation of counterfeit e-waste certificates** — directly relevant to our verification/badge design (Section 5.3)
 - Recommends formal-informal sector collaboration rather than exclusion as the effective mitigation path
3. **“Modeling of barriers in integrating informal E-waste collectors into India’s formal E-waste management system”** (ResearchGate) — in-depth interviews with 23 informal e-waste collectors across high-density Delhi markets (Karol Bagh, Lajpat Nagar)
 - Identifies **“lack of support system”** and **“inconvenience”** as primary barriers against formal-channel participation — directly informing our design principle of minimizing collector-side friction (single photo/voice listing flow, no complex paperwork)
 - Identifies **“salvage value”** (i.e., fair pricing) as a primary motivator *for* participation — informs the priority given to the price-discovery/trend feature
4. **“Comparative Analysis of Formal and Informal E-Waste**

- Disposal Methods in Gurgaon”** (*Clean-World Environmental Journal*)
- Quantifies that informal processing in Gurgaon handles >95% of e-waste and produces measurably higher air/water/soil contamination than formal “wash and drain” recycling methods — used as supporting evidence for the platform’s core value proposition in the pitch narrative
5. **“Understanding the gap between formal and informal e-waste recycling facilities in India”** (ResearchGate)
 - Formal channels currently process only an estimated ~15% of India’s total e-waste, despite 300+ registered recyclers with multi-million-tonne annual capacity
 - Identifies institutional/coordination gaps (not capacity gaps) as the primary bottleneck — reinforces this platform’s framing as an information/matching problem rather than a processing-capacity problem
 6. **Electronic waste management in Northern India: regional case study of Chandigarh** (*Journal of Material Cycles and Waste Management*, Springer, 2025)
 - Direct kabadiwala interview transcripts describing existing informal hierarchy (kabadiwalas sub-contracting to “feriwalas” for household-level collection)
 - Documents that some informal collectors already attempt informal collaboration with formal recyclers via shared affidavits/licenses — validates that a formal bridge mechanism (this platform) addresses a demand that already exists organically, rather than requiring behavior change from zero
 7. **Nowakowski & Pamuła, “Application of deep learning object classifier to improve e-waste collection planning”** — foundational reference for photo-based e-waste classification
 - Demonstrated a mobile-platform-based system where images are captured on-device, uploaded, and classified server-side using CNN, reporting 90–97% accuracy even on a small training set (180 images across 3 classes)
 - Directly informs our client-uploads/server-infers architecture for the classification feature (Section 7.1), and establishes that meaningful accuracy is achievable even with a modest, hackathon-scale labeled dataset
 8. **“Classification of Waste Materials using CNN Based on Transfer Learning”** (ACM Digital Library)
 - Benchmarks multiple pretrained backbones (MobileNet, ResNet50, Xception, InceptionV3, DenseNet201) fine-tuned via transfer learning on the Stanford TrashNet dataset (~3,242 images, 6–7 classes)
 - Directly supports the transfer-learning approach specified in Section 7.1, and the decision to supplement limited self-collected images with an existing public dataset structure rather than training from scratch
 9. **“Recyclable Waste Image Recognition Based on Deep Learning”** (Hassan & Gouda) and **“WasteInNet”** (ScienceDirect)
 - Report CNN-based classification accuracy up to 95.9–97% on recyclable/e-waste categories using MobileNet/Xception hybrid architectures — used to benchmark realistic accuracy expectations for our own fine-tuned model

3.2 Data Sources Identified for Platform Use

Data Type	Source	Use in Platform
CPCB EPR registration format & registry structure	E-Waste (Management) Rules, 2022; CPCB EPR Portal FAQ	Regex-based registration number validation (Section 5.2)
	Self-collected field photographs + public	Fine-tuning the

Material image training data	datasets (Stanford TrashNet, Kaggle/Roboflow e-waste sets)	classification model (Section 7.1)
Fair-price benchmarks	Field interviews with local collectors/aggregators (primary data collection, per official PS requirement)	Seeding the initial price-discovery dataset (Section 7.2)
Collector behavior & motivators	Sengupta et al. (2023); Delhi barrier-modeling study	Informs UX flow prioritization — minimal friction, fair pricing prominently displayed
Regulatory/compliance gap evidence	Chandigarh case study (counterfeit certificates finding)	Justifies mandatory verification gate design (Section 5)

3.3 How Research Findings Map to Design Decisions

- **Counterfeit certificate circulation (Chandigarh study) → Verification Gate (Section 5.2):** since fraudulent authorization documentation is a documented, real problem in the field (not a hypothetical risk), the platform’s manual verification queue and regex-based registration validation are direct responses to an evidenced failure mode, not speculative design.
- **Low formal-channel penetration (~15%, per gap-analysis study) → Platform framing:** confirms the problem is one of coordination/incentive, not recycling capacity — supports positioning the platform as a matching/information layer rather than attempting to build new processing infrastructure.
- **Collector friction & inconvenience as top barrier (Delhi interview study) → Feature prioritization:** directly justifies why voice-based listing and single-photo classification (low-effort input methods) were prioritized over more data-heavy onboarding flows.
- **Fair pricing as primary motivator (same study) → Price trend feature priority:** price transparency is treated as a core feature, not a “nice-to-have,” because the literature identifies it as the strongest lever for behavior change.
- **Existing organic informal-formal collaboration attempts (Chandigarh field interviews) → Product validation:** confirms this platform formalizes and scales a bridging behavior collectors already attempt informally, rather than requiring an entirely new behavior.
- **CNN transfer-learning accuracy benchmarks (90-97% across cited studies, even on small training sets) → Technical feasibility confidence:** supports the technical viability of Section 7.1’s approach within a hackathon-scale dataset, rather than requiring a large-scale proprietary dataset to be credible.

4. Scope & Approach

The platform implements a **three-tier role-based architecture**, extending beyond the base problem statement’s two-party model (Collector ↔ Recycler) to include the origination layer (Seller), closing the loop end-to-end:

Role	Description	Regulatory Status
Seller	Household/individual	No registration required

	originating scrap	
Collector	Informal aggregator (kabadiwala)	No mandatory EPR registration (facilitation role only, per 2022 Rules)
Recycler	CPCB-authorized processing facility	Mandatory EPR registration under 2022 Rules

For the current submission phase, the platform is scoped as a **native mobile application** (not a web app), a requirement driven directly by the official brief's offline-tolerance and low-end-device optimization mandate — native storage and background sync capabilities are materially more reliable than what a web app can offer for this use case.

5. User Roles & Access Control

5.1 Authentication & Onboarding

- Phone number + OTP-based authentication (primary identity)
- Role selection at signup: seller / collector / recycler

5.2 Verification Gate (Access Control Logic)

Collector and Recycler roles are subject to a **mandatory pre-dashboard verification gate**:

verification_status ENUM:

- PENDING → dashboard renders in read-only mode, all write actions disabled
- COLLECTOR_VERIFIED → basic KYC confirmed (name, shop, location); no EPR requirement
- RECYCLER_VERIFIED → KYC + EPR registration number + certificate validated
- REJECTED → account flagged, access denied

Verification requirements by role: - Collector: name, shop/operating name, geolocation, phone (OTP), operating materials - Recycler: above + CPCB EPR registration number + certificate upload

Validation approach (no third-party verification API): - Registration number format validated via regex pattern matching against known CPCB numbering conventions (client + server-side) - Certificate document stored and routed to a manual approval queue for the current build phase (simulates a compliance officer review step; production version would integrate directly with the CPCB registry) - Verification itself requires connectivity (one-time event at onboarding); once verified, status is cached locally on-device so badge display and dashboard access remain available offline thereafter

5.3 Trust Badge System

Rendered conditionally based on verification_status, surfaced wherever a user profile appears (search results, price board, match results):

-  **Authorized Recycler** — RECYCLER_VERIFIED
-  **Collector** — COLLECTOR_VERIFIED (no registration required by regulation, badge communicates role transparently rather than compliance status)

-  **Unverified** — PENDING / REJECTED

This directly operationalizes the informational gap identified in the problem statement: collectors currently have no reliable mechanism to distinguish compliant recyclers from unregistered entities.

6. Core User Flow

1. Seller captures scrap item via photo or voice input → structured listing created (works with or without connectivity)
 2. Platform surfaces ranked nearby Collectors (by offered price, distance, verification status) — from cached local data if offline, live data if online
 3. Seller and Collector coordinate directly via displayed contact details and agree on pickup
 4. Handover executed; both parties confirm via QR-code exchange → transaction record created locally, synced to server when connectivity is available
 5. Collector aggregates lots, lists to nearby verified Recyclers
 6. Recycler confirms intake via QR scan → completes chain-of-custody record
 7. Both Collector and Recycler earnings ledgers update accordingly, syncing across devices once online
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7. Feature Specifications

7.1 Photo-Based Material Classification & Condition Grading

Function: User captures an image of scrap material → system returns material category (PCB, CRT, LCD panel, cable, battery, motor, mixed plastic, etc.) and a condition grade (good/moderate/poor) → generates an estimated value.

Technical approach: - Base architecture: **MobileNetV2** or **EfficientNet-Lite**, selected for low-latency inference and mobile-appropriate model size - **Transfer learning strategy:** initialize with ImageNet-pretrained weights, fine-tune final layers on a labeled scrap-material dataset (target: 200-500 labeled images per category, supplemented with public e-waste datasets from Kaggle/Roboflow to offset limited self-collected data) - Condition grading implemented as a secondary lightweight classification head, or a simplified heuristic (image contrast/discoloration proxy) as a fallback if training data for condition labels is insufficient within the build window - **Model exported to TensorFlow Lite (.tflite) and bundled with the app for fully on-device inference** — required since classification must work without connectivity; no server round-trip - **No third-party vision API used** — model is self-trained, self-hosted, and runs entirely on-device

Known constraint: classification accuracy is directly bounded by training data volume; demo reliability should be validated against a curated evaluation set prior to live presentation. On-device inference also requires monitoring model file size to stay within the low-end-device optimization target (Section 8.1).

7.2 Price Discovery & Trend Analysis

Function: Time-series view of material prices by category/location, derived from platform transaction history.

Technical approach: - Historical price points persisted per transaction (material category, location, timestamp, quoted price, final price) - Trend computation via **moving average / linear regression** (scikit-learn) — deliberately lightweight, no deep learning required for this feature - Latest synced price data cached locally so the trend chart still renders when offline; recalculated/refreshed on next sync - For demonstration purposes, seeded with synthetic historical data to populate meaningful trend visualization ahead of sufficient real transaction volume - Rendered client-side via Recharts/Chart.js

7.3 Voice-Based Listing Creation

Function: User speaks item details in Hindi/Marathi; system transcribes and auto-populates structured listing fields (material type, approximate weight).

Technical approach (current phase): - Speech-to-text via **Vosk** (offline-capable, supports Hindi/Marathi acoustic models), bundled on-device — already inherently offline-compatible, no architecture change needed for this feature - Post-transcription parsing via a constrained **rule-based / keyword-matching extraction layer** rather than a general-purpose NLP model — appropriate given the narrow, closed vocabulary domain (material names, units, numerals), and runs on-device alongside the STT engine - Optional extension path: fine-tuned **spaCy NER model** trained on a small custom-labeled corpus of scrap-related utterances, if entity extraction accuracy from rule-based matching proves insufficient - *Note: an API-based STT/NLP upgrade path (e.g., Whisper) is identified as a future iteration and is intentionally excluded from the current build to keep the submission API-independent.*

7.4 QR-Code Verified Handover Record

Function: Generates a scannable, verifiable digital record at the point of material handover.

Technical approach: - QR payload: JSON-encoded transaction metadata (lot/reference ID, material category, weight, quoted/final price, GPS coordinates, timestamp) - Generation: qrcode library (client-side, fully offline) - Verification: receiving party scans via device camera (react-native-camera or equivalent), decodes and cross-references against the locally stored transaction record, marks status as confirmed - No AI/ML component, no connectivity dependency — pure on-device data-encoding and validation logic

7.5 Recycler Authorization Badge

Covered under Section 5.3 — implemented as conditional UI rendering against `verification_status`, no independent technical component.

8. Architecture Decisions

8.1 Connectivity Model — Offline-Tolerant (Not Offline-Only)

Per the official problem statement's requirement that the platform "operate in low-connectivity environments" with "core activities completed offline and synchronized when connectivity becomes available," the platform is built **offline-tolerant**: it functions fully

when online, and gracefully continues functioning — not fails — when offline, rather than requiring a permanent internet connection or being restricted to offline-only use.

Behavior by connectivity state: - **Online:** all actions (listing creation, price lookups, matching, badge verification, sync) occur in real time against the server - **Offline:** core actions remain available — photo classification (on-device model), voice listing (on-device Vosk), QR generation/scanning, and local listing creation — all persisted to local device storage - **Reconnection:** locally queued data (listings, transactions, ledger updates) automatically syncs to the server in the background; server-side updates (new price data, newly verified recyclers) pull down to the device

This is a deliberate middle ground: the app is not built assuming permanent connectivity (which the official brief explicitly rules out), nor is it built as a fully disconnected, offline-only product (which would be unrealistic given collectors do generally have basic mobile network access). It optimizes for the common case (online) while remaining functional in the stated worst case (offline).

8.2 Data Architecture

Dual-layer storage: local on-device storage (**SQLite**) for offline-first persistence, synced to a central server database (**PostgreSQL**) whenever connectivity is available. Sync uses a queue-based approach — locally created/modified records are marked pending-sync and pushed in order once online; conflict resolution follows a last-write-wins policy for the current build phase.

8.3 On-Device AI Optimization

Given the offline requirement, AI/ML components that were previously considered for backend-hosted inference are instead run **on-device:** - Vision classification model exported to **TensorFlow Lite (.tflite)** — a compressed, mobile-optimized model format - Speech-to-text via Vosk, already offline-native by design - Both are selected and sized specifically to remain performant on **entry-level Android hardware**, per the official brief's low-resource-device requirement — model size and memory footprint are treated as first-class constraints, not just accuracy

8.4 API Independence

All AI/ML functionality (vision classification, speech-to-text, price trend modeling) is implemented via self-trained/self-hosted models rather than third-party paid APIs, consistent with the project's current build constraints. This is treated as a deliberate architectural choice through the current submission phase, with defined upgrade paths (e.g., API-based STT) identified for future iterations post-selection.

9. Technology Stack

Layer	Technology	Notes
Client	React Native	Native mobile app, required for reliable offline storage/sync
Local Storage	SQLite	On-device persistence layer
Backend	Node.js/Express or Python/FastAPI	REST sync endpoints

Server Database	PostgreSQL	Central store, synced from device-local data
Vision Model	MobileNetV2 / EfficientNet-Lite → TensorFlow Lite	Fine-tuned, compiled for on-device inference
Speech-to-Text	Vosk	Offline-native, Hindi/Marathi support, runs on-device
NLP Extraction	Rule-based / spaCy NER	Constrained-vocabulary entity extraction, on-device
Trend Analysis	scikit-learn	Regression / moving average, server-side with local cache
QR Generation/Scan	qrcode (client) + camera library	Fully offline, no external API

10. Datasets Required (per official PS specification)

- **Material Dataset:** category, sub-category, description, image reference, weight, condition, source type, estimated value
 - **Price Dataset:** category, location, date/time, buying price, quoted price, unit, recycler/aggregator, historical trend
 - **Recycler Dataset:** name, facility location, materials accepted, authorization status/details, contact, offered rate, pickup availability, service area
 - **Transaction Dataset:** lot ID, collector ID, material category, weight, quoted/final price, recycler ID, locations, timestamps, payment/transaction status
 - **Traceability Dataset:** lot ID, photographs, weight, timestamp, GPS, handover reference, recycler confirmation
 - **Collector Dataset:** minimal profile — collector ID, preferred language, operating location, transaction/earnings history (no unnecessary PII collection)
 - **AI/ML Training Dataset:** labeled material images with category, weight, price, location, transaction metadata — source, quality, size, and limitations to be explicitly documented per official PS requirement
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11. Deliverables (per official PS expected outcome)

- Working mobile application (Seller, Collector, Recycler interfaces)
 - Structured datasets as specified in Section 9
 - Field research: minimum 2 real scrap collectors/aggregators consulted
 - Live usability demonstration
 - Unit-economics assessment: comparative analysis of a collector's current earnings vs. projected earnings via the platform, with an explanation of platform sustainability
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12. Open Items for Team Confirmation

- Final labeled-image dataset size/sourcing plan for Section 7.1
- Manual verification queue owner (who reviews Recycler certificate uploads during demo phase)
- Field research logistics — identifying and scheduling the 2 required collector interviews